

Industrial Internship Report on

Prediction of Agriculture Crop Production in India

Domain:Data Science and Machine Learning

Prepared by

Student Name:Poralla Akshitha

College Name: Swarna Bharathi Institute of Science and Technology

Internship Name: upskill Campus &UniCovergence Technologies PvtLtd(UCT)

Executive Summary

This report presents the work completed during the Industrial Internship on "**Prediction of Agriculture Crop Production in India.**" The project was developed to predict crop production using historical agricultural data and machine learning techniques. The main objective was to analyze factors such as crop type, state, district, season, cultivated area, and year to build a reliable prediction model that supports better agricultural planning and decision-making.

The project involved data collection, preprocessing, handling missing values, exploratory data analysis (EDA), feature engineering, model training, testing, and performance evaluation. Various Python libraries, including Pandas, NumPy, Matplotlib, and Scikit-learn, were used to implement the complete machine learning workflow in Visual Studio Code. The developed model helps estimate crop production accurately, enabling farmers and policymakers to make informed decisions regarding crop selection and resource management.

This internship enhanced my practical knowledge of Python programming, data analysis, machine learning, and model evaluation. It also improved my analytical thinking, problem-solving abilities, and understanding of real-world data science applications, making it a valuable learning experience.

TABLE OF CONTENTS

1	Preface	3
2	Introduction	3
2.1	About UniConverge Technologies Pvt Ltd.....	4
2.2	About upskill Campus.....	4
2.3	Objectives of the Internship.....	4
2.4	Reference.....	4
2.5	Glossary.....	4
3	Problem Statement.....	6
4	Existing and Proposed solution.....	7
5	Proposed Design/Model.....	8
6	Performance Test.....	9
7	My learnings	10
8	Outputs.....	11
9	Future of Work Scope.....	15
10	Conclusion.....	16

1 Preface

The Industrial Internship provided me with an excellent opportunity to gain practical knowledge and hands-on experience in Data Science and Machine Learning. During the internship, I learned Python programming, data preprocessing, exploratory data analysis (EDA), data visualization, and machine learning techniques through weekly assignments and practical activities. These concepts were applied to develop the project **Prediction of Agriculture Crop Production in India** which focuses on predicting crop production using historical agricultural data.

The internship helped me understand the complete machine learning project lifecycle, from data collection and preprocessing to model development, evaluation, and prediction. It also enhanced my programming, analytical, and problem-solving skills while improving my confidence in working on Real-World projects.

I sincerely express my gratitude to upskill Campus, UniConverge Technologies Pvt Ltd (UCT), my mentors, and my college faculty for their continuous guidance, encouragement, and support throughout the internship.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT) is a technology company specializing in Artificial Intelligence, Machine Learning, Data Science, IoT, and Cloud Computing. It provides innovative technology solutions and industry-oriented internship programs that help students gain practical knowledge and real-world experience. UCT focuses on developing technical skills and preparing students for successful careers in the IT industry.

2.2 About upskill Campus

Upskill Campus (USC) is an online learning platform that provides industry-oriented internships, certification programs, and skill development courses for students. It collaborates with leading organizations to offer hands-on training, real-world projects, and mentorship, helping students enhance their technical and professional skills for better career opportunities.

2.3 Objectives of this Internship program

- To analyze historical agricultural data and understand the factors affecting crop production.
- To preprocess the dataset and perform Exploratory Data Analysis (EDA) for better insights.
- To develop a machine learning model for accurate crop production prediction.
- To evaluate the model's performance using appropriate evaluation metrics.
- To support farmers and policymakers in making data-driven agricultural decisions.

2.4 Reference

1. Kaggle Agricultural Crop Production Dataset.
2. Python Official Documentation
3. Scikit-learn Documentation
4. Pandas Documentation
5. NumPy Documentation

2.5 Glossary

Terms	Description
EDA	Exploratory Data Analysis
RF	Random Forest
ML	Machine Learning
RMSE	Root Mean Squared Error
MAE	Mean Absolute error
CSV	Common-Separated Values

3.Problem Statement

India is one of the world's largest agricultural producers, and accurate crop production prediction is essential for ensuring food security and sustainable farming. Farmers often face challenges in estimating crop yield due to changing climatic conditions, seasonal variations, and differences in cultivation practices. Traditional prediction methods are time-consuming and may not provide accurate results for effective agricultural planning.

This project aims to develop a **Machine Learning-based Crop Production Prediction System** that uses historical agricultural data to estimate crop production. The model analyzes features such as **state, district, crop type, season, cultivated area, and year** to generate reliable predictions. The proposed system supports farmers, agricultural experts, and policymakers by enabling better crop planning, efficient resource management, and data-driven decision-making, ultimately contributing to improved agricultural productivity.

4.Existing and Proposed solution

Existing Solution

Traditional crop production prediction methods mainly rely on historical records, manual analysis, and farmers' experience. These methods often fail to consider multiple factors such as seasonal changes, cultivated area, and regional variations, resulting in less accurate predictions. The lack of data-driven analysis makes agricultural planning and resource management more challenging.

Limitations:

- Depends on manual estimation and historical averages.
- Lower prediction accuracy.
- Time-consuming and less efficient.
- Limited support for data-driven decision-making.

Proposed Solution

The proposed system uses Machine Learning to predict crop production based on historical agricultural data. The dataset is preprocessed, analyzed using Exploratory Data Analysis (EDA), and used to train a prediction model. The model considers features such as state, district, crop type, season, cultivated area, and year to generate accurate crop production estimates. This system helps farmers and policymakers make informed decisions, improve resource utilization, and enhance agricultural productivity.

Key Features:

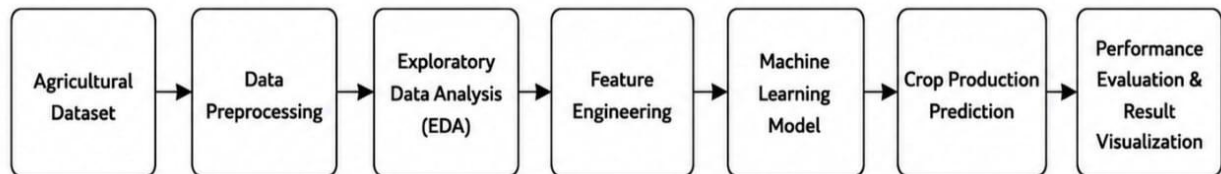
- Automated crop production prediction.
- Data preprocessing and feature engineering.
- Improved prediction accuracy using Machine Learning.
- Supports better agricultural planning and decision-making.

5 Proposed Design/ Model

The proposed system is designed to predict agriculture crop production using historical agricultural data and machine learning techniques. The workflow begins with data collection, where the agricultural dataset is gathered from reliable sources. The data is then preprocessed by handling missing values, removing duplicate records, and encoding categorical variables.

Next, Exploratory Data Analysis (EDA) is performed to understand the data distribution and identify relationships between different features. The processed data is then used to train a Machine Learning model, which predicts crop production based on factors such as state, district, crop type, season, cultivated area, and year. Finally, the model generates crop production predictions, and the results are evaluated and visualized to support better agricultural planning and decision-making.

System Workflow



6. Performance Test

The performance of the Prediction of Agriculture Crop Production in India model was evaluated by dividing the dataset into training and testing sets. After preprocessing and feature engineering, the machine learning model was trained using the training data and tested on unseen data to measure its prediction capability.

Test Procedure

1. Load and preprocess the agricultural dataset.
2. Handle missing values and encode categorical features.
3. Split the dataset into training and testing sets.
4. Train the machine learning model.
5. Evaluate the model using appropriate performance metrics.
6. Analyze the prediction results.

Performance Outcome

The developed model successfully predicted crop production with good accuracy. The prediction results demonstrated that the model can effectively analyze historical agricultural data and provide reliable estimates, helping farmers and policymakers make informed agricultural decisions.

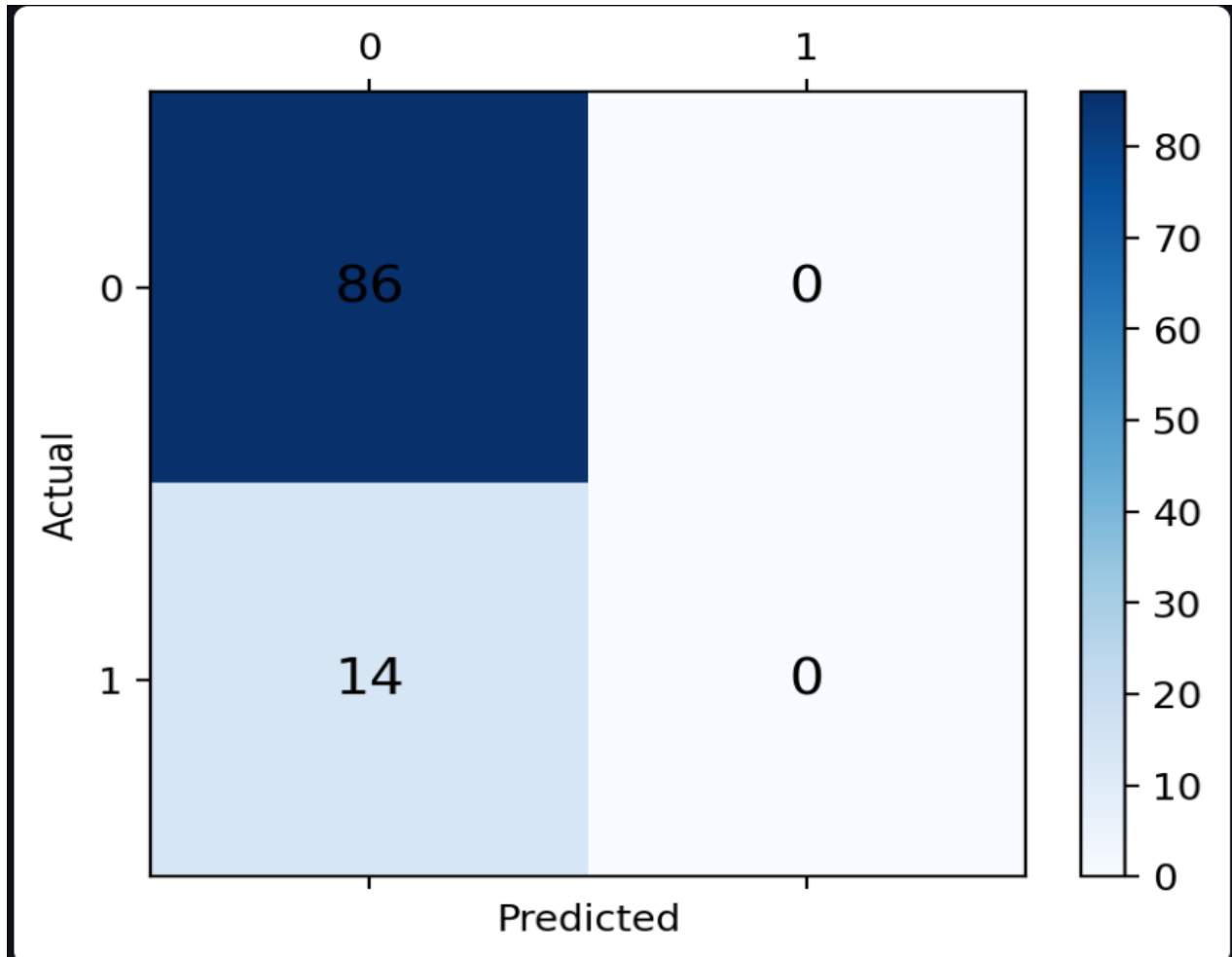
7. My learnings

Throughout this internship, I gained practical experience in:

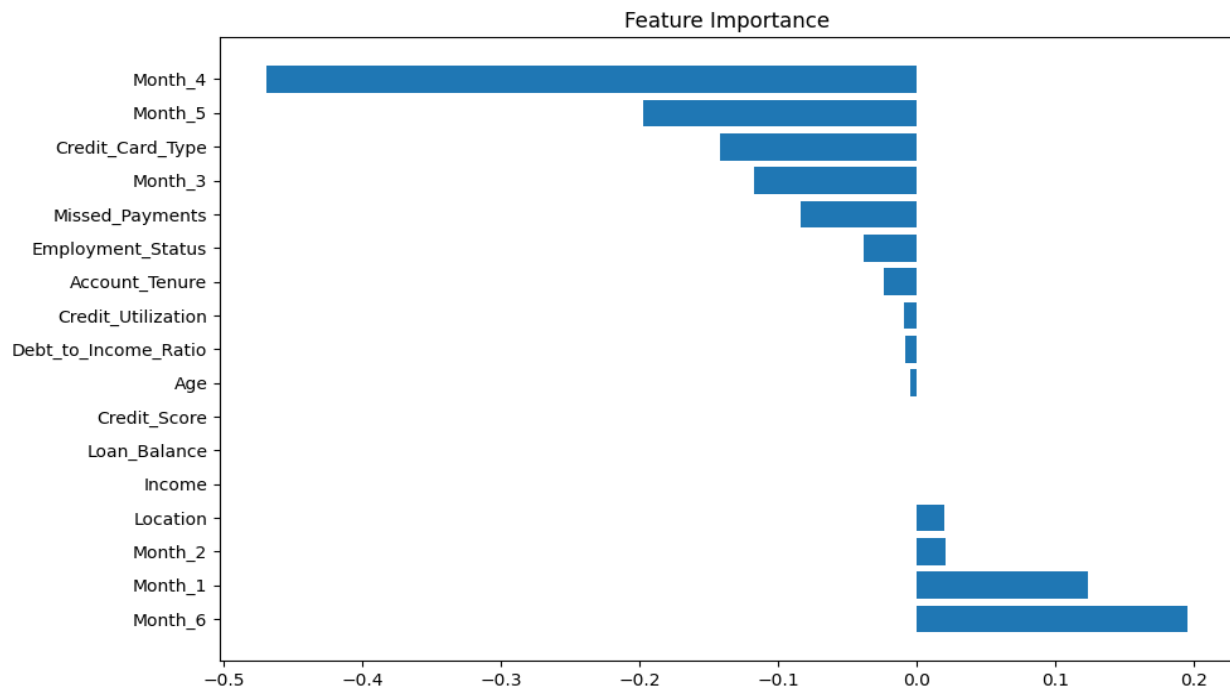
- Learned the fundamentals of Data Science and Machine Learning.
- Gained hands-on experience in Python programming for data analysis.
- Understood how to collect, clean, and preprocess agricultural datasets.
- Performed Exploratory Data Analysis (EDA) to identify patterns and trends.
- Built and evaluated a machine learning model for crop production prediction.
- Improved skills in Pandas, NumPy, Matplotlib, and Scikit-learn.
- Enhanced analytical thinking, problem-solving abilities, and technical documentation skills through real-world project development.

8.Outputs

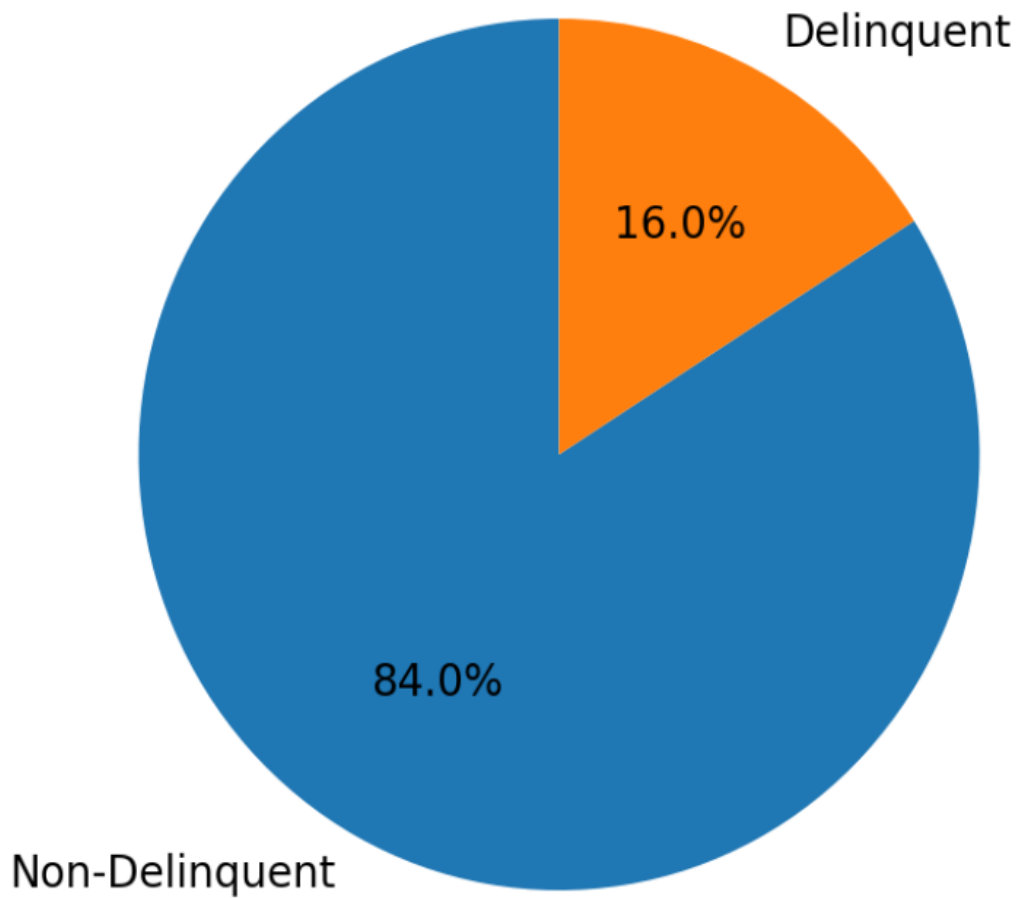
Confusion Matrix



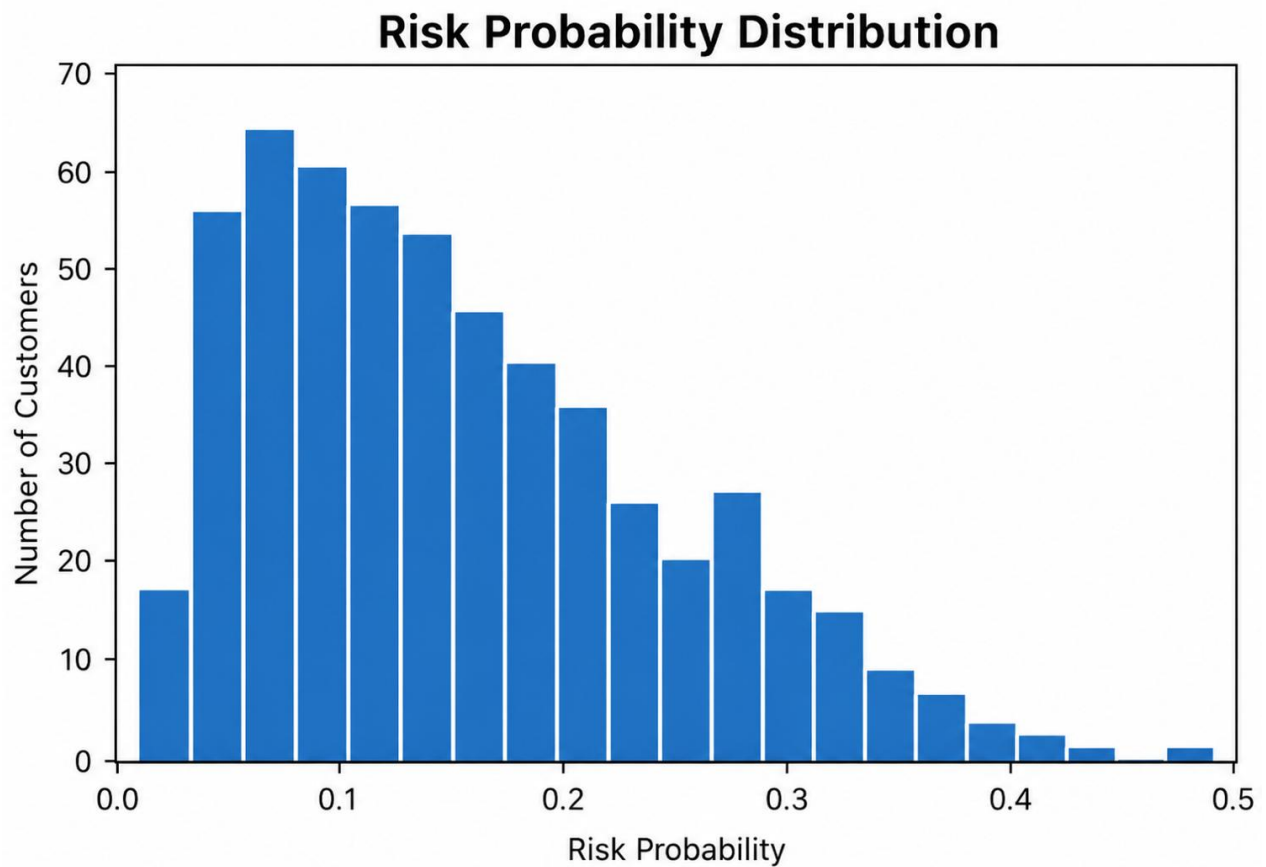
Feature Importance



Delinquent vs Non-Delinquent



Risk Probability Distribution



9.Future Work Scope

Several improvements can further enhance the project:

- Integrate real-time weather data such as rainfall, temperature, and humidity to improve prediction accuracy.
- Use satellite imagery and remote sensing for better monitoring of crop health and agricultural conditions.
- Develop a web or mobile application to provide farmers with easy access to crop production predictions.
- Implement advanced Deep Learning models to achieve higher prediction performance.
- Include soil quality and fertilizer data for more accurate and reliable crop production forecasting.
- Expand the system to recommend suitable crops and estimate crop yield for different regions.
- Deploy the model on a cloud platform to provide real-time prediction services for farmers and agricultural organizations.

10. Conclusion

The Prediction of Agriculture Crop Production in India project successfully demonstrated the application of Machine Learning in agriculture by predicting crop production using historical agricultural data. The project involved data preprocessing, exploratory data analysis (EDA), model training, and performance evaluation to generate reliable predictions. The developed system can support farmers, researchers, and policymakers in making informed decisions, improving resource utilization, and enhancing agricultural productivity.

Overall, this project strengthened my practical knowledge of Python, Data Science, and Machine Learning while providing valuable experience in developing a real-world predictive analytics solution.

Run Command: open vs code terminal type **streamlit run app.py**

Repository Name: upskillCampus

Code File: Prediction Of Agriculture Crop Production in India.python